

Epistasis limits stepwise strain engineering: a complete three-factor test in fatty acid production (13/15 words)

Michael Volk^{1,2,3*} and Huimin Zhao^{1,2,3*}

¹DOE Center for Advanced Bioenergy and Bioproducts Innovation, University of Illinois Urbana-Champaign, Urbana, IL, 61801, USA.

²Department of Chemical and Biomolecular Engineering, University of Illinois Urbana-Champaign, Urbana, IL, 61801, USA.

³Carl R. Woese Institute for Genomic Biology, University of Illinois Urbana-Champaign, Urbana, IL, 61801, USA.

*Corresponding author(s). E-mail(s): mjvolk3@illinois.edu; zhao5@illinois.edu;

Abstract (≤ 150 words)

Strain engineering proceeds one edit at a time, keeping what improves and discarding what does not. The procedure assumes that an improving combination can be reached through improving intermediates. We test that assumption on a complete three-factor design: ten transcription factors deleted in every single, double and triple combination in a free-fatty-acid-producing *Saccharomyces cerevisiae* strain, with all 720 deletion orders measured rather than inferred. Interaction is the rule. Of 119 testable triples, 86 deviate from the multiplicative null at a 5% false discovery rate, and the median deviation moves from **+0.40** at two deletions to **−0.72** at three. Fifty-one triples exceed the base strain, yet only two sit at the end of a route whose every intermediate also improves, and the shallowest route falls a median 31% below the starting titer before recovering. None of the seven interaction networks tested is enriched among the interacting triples.

Keywords: metabolic engineering, epistasis, gene interactions, strain design, combinatorial deletion, *Saccharomyces cerevisiae*

Status: draft of a separate publication, not a working note. This is intended for submission as a standalone perspective once the free fatty acid study it re-analyzes is published, so it deliberately tracks the house style of the TorchCell manuscript (paper/nature-biotech/editing.pdf): same sn-jnl class, same sn-nature bibliography style, same print-approximate editing geometry, same section status chips. Keep it that way, and when a convention changes in paper/nature-biotech, change it here too.

Two things are deliberately unfinished. Authorship is a placeholder and must include the authors of the source study. Citations are not wired: the document has no Zotero collection, so prior work is named in prose rather than through \cite, and a bibliography needs that collection created first.

✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

Contents

1	Introduction ✗	3
2	Results ✗	3
	A complete three-factor design over ten transcription factors	3
	Interaction is the rule, and the third deletion is where it bites	3
	Improving combinations are not reachable one deletion at a time	4
	What counts as an interaction depends on the null and on the scale	5
	No interaction network on hand predicts which combinations interact	6
3	Discussion ✗	7
	What this design can and cannot support	8
	What would settle it	8
4	Methods ✗	8
	Data and normalization	8
	Epistasis models	8
	Uncertainty and multiple testing	9
	Path enumeration and monotonicity	9
	Interaction graph enrichment	9
	Figures	9
	Code and data availability	9

1 Introduction ✕

A strain is improved by editing it, measuring it, keeping the edits that helped, and editing again. Every round of that loop is a decision made on the strain in hand, and a round that produced nothing better than its parent is a round whose output is discarded. The procedure is the design-build-test-learn cycle in its most common form, and its cost scales with the number of rounds rather than with the number of edits eventually carried.

The procedure is efficient exactly when the effects of edits compose. If a deletion that raises titer on its own keeps raising titer in the presence of other deletions, then a greedy walk reaches a good combination, and the number of strains that must be built grows linearly with the number of edits rather than exponentially. The assumption is rarely stated because it is rarely testable: a campaign that builds a few hundred strains out of a combinatorial space of millions cannot see the combinations it did not build, and cannot tell a combination that is genuinely bad from one it never reached.

Epistasis is the name for the failure of that assumption. In its quantitative form it is the deviation of a combination’s phenotype from what the phenotypes of its parts predict, and it is measured against a stated null. Systematic genetic interaction mapping in *Saccharomyces cerevisiae* established both the scale of the phenomenon on growth rate and the estimator now used for it: a digenic interaction score defined as the departure of a double mutant’s fitness from the product of the two single-mutant fitnesses, and a trigenic score that removes the digenic contributions before asking what the third deletion adds. Those maps were built on fitness in rich medium. Whether their conclusions transfer to a production phenotype is not obvious, because fitness and titer are under different selection, are measured on different instruments, and need not share a null.

What has been missing for a production phenotype is a design in which every combination and every construction order is present. Without one, a claim that a combination was unreachable is indistinguishable from a claim that nobody happened to build it. The free-fatty-acid dataset re-analyzed here supplies exactly that: ten transcription factors deleted in every single, double and triple combination in one producing background, which makes all 720 orders of deleting three genes available as measurements rather than as inferences. The design permits a question that a partial screen cannot ask. Not how many combinations interact, which any screen can estimate, but how many good combinations a stepwise campaign could have found.

2 Results ✕

A complete three-factor design over ten transcription factors

The dataset is a combinatorial deletion series in a free-fatty-acid-producing *S. cerevisiae* strain. Ten transcription factors (FKH1, GCN5, MED4, OPI1, RFX1, RGR1, RPD3, SPT3, TFC7 and YAP6) were deleted in every combination of one, two and three genes, giving 10 single, 45 double and 120 triple deletion strains alongside the undeleted base strain. The base strain is itself an engineered chassis, an FFA-overproducing *S. cerevisiae* carrying *pox1Δ faa1Δ faa4Δ*, and the ten transcription factor deletions are additional to those three. Every titer below is relative to that chassis, so a triple deletion strain carries six deletions in total. Six quantities were measured per strain: the titers of myristate (C14:0), palmitate (C16:0), stearate (C18:0), palmitoleate (C16:1) and oleate (C18:1), and their sum, which we refer to as total titer. All titers below are normalized to the base strain, so a value of 1 is no change and 2 is a doubling.

Completeness is what makes the design usable for the question at hand. Because every sub-combination of every triple was built, each triple can be reached by six distinct deletion orders, and the intermediate strain at each step of each order was measured rather than modeled. The design therefore contains 720 fully measured construction routes.

Only one of the ten single deletions raises titer at all. Deleting GCN5 gives 1.06×; the other nine lose ground, and deleting RPD3 costs nearly half the titer at 0.52× (Fig. 2a). Combinations behave very differently: 29 of the 45 doubles and 51 of the 120 triples exceed the base strain, and the best strain in the design is a triple at 2.05×. The gap between what single deletions do and what combinations do is the first sign that the effects are not composing.

Interaction is the rule, and the third deletion is where it bites

We scored each combination against a multiplicative null, in which the expected phenotype of a combination is the product of its parts’ phenotypes. For a double deletion the digenic score is $\varepsilon_{ij} = f_{ij} - f_i f_j$. For a triple, the trigenic score removes the digenic contributions before asking what remains:

$$\tau_{ijk} = f_{ijk} - f_{ij}f_k - f_{ik}f_j - f_{jk}f_i + 2f_i f_j f_k. \quad (1)$$

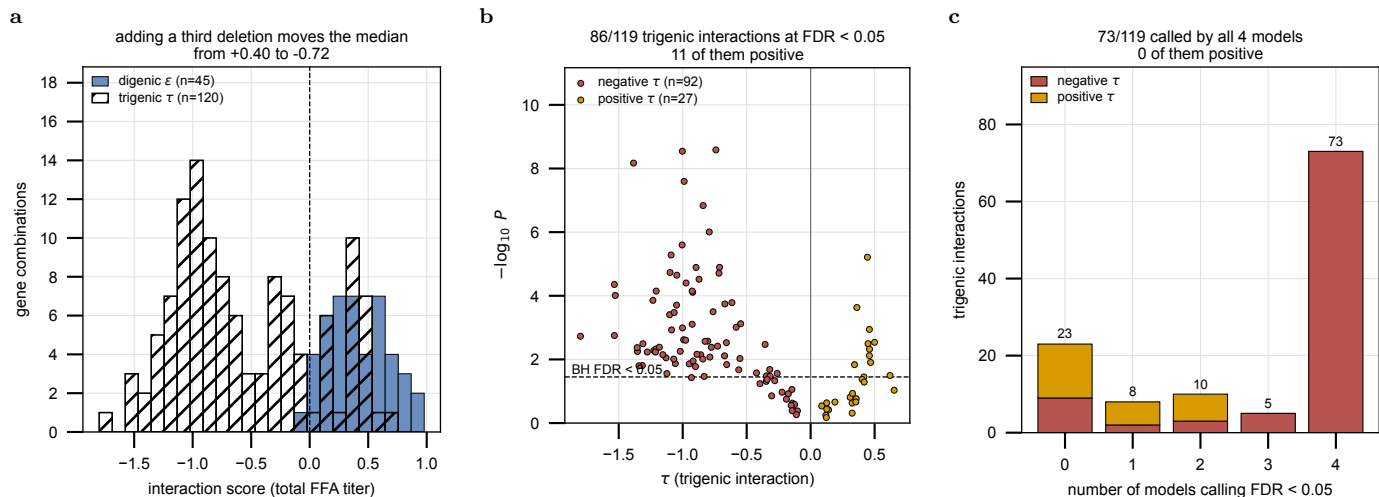


Fig. 1 Interaction is the rule on total fatty acid titer, and the third deletion is where it bites. **a**, Interaction scores against the multiplicative null for all 45 double (fill) and 120 triple (hatched) deletions. The median moves from +0.40 at two deletions to -0.72 at three. **b**, Trigenic interaction score τ against evidence for the 119 testable triples under the multiplicative model. The dashed line marks the largest P that Benjamini-Hochberg rejects at a 5% false discovery rate over the 119 tests of this readout ($P = 0.036$); 86 triples clear it and 11 of those have $\tau > 0$. **c**, How many of the four epistasis models call each triple significant, split by the sign of τ . Seventy-three triples are called by all four with the sign agreeing, and none of the 73 is positive.

This is the trigenic score of the yeast genetic interaction maps, applied to titer in place of fitness. Standard errors propagate through Eq. 1 by the delta method, and each t statistic is referred to the Welch-Satterthwaite effective degrees of freedom of that linear combination, which has a median of 4.28 across the 714 testable trigenic scores.

Deviations from the null are large and are mostly negative at three deletions. The median digenic score is +0.40 and the median trigenic score is -0.72 (Fig. 1a), so adding a third deletion does not damp the interaction, it reverses its typical direction. Of the 119 triples with a testable standard error, 86 depart from the null at a 5% false discovery rate computed over the 119 tests of this readout, and 11 of those 86 depart upward (Fig. 1b).

Improving combinations are not reachable one deletion at a time

An interaction score says a combination is surprising. It does not say whether a campaign could have found the combination, and that is the question a strain engineer is actually asking. We therefore scored each of the 720 measured routes for monotonicity: a route is monotone when every deletion in it leaves the strain strictly more productive than the strain before it, which is the condition under which a build-and-screen campaign that discards non-improving rounds would follow the route to its end.

Two of the 720 routes are monotone. Relaxing the criterion to allow a step whose loss is within one standard error raises the count to 11 routes across 9 triples, so the result is not an artifact of measurement noise sitting just the wrong side of zero. Counting by destination rather than by route: 51 of the 120 triples beat the base strain, and 2 of those 51 sit at the end of a monotone route (Fig. 2c). The other 49 improvements exist in the design and are invisible to a stepwise campaign.

The obstacle is a valley rather than a plateau. For each triple we took its shallowest route, meaning the one whose worst intermediate is highest, and measured how far that worst intermediate falls below the base strain. The median shallowest route dips 31% below the starting titer, and the deepest dips 48%. A campaign built to discard anything worse than its parent will discard the route at the first step.

Simulating the campaign directly gives the same answer in one number. Starting from the base strain and taking the best strictly improving single deletion each round, with full visibility of every candidate at every round, the campaign deletes GCN5 (1.06 $\times$), then TFC7 (1.80 $\times$), and then stops, because all eight remaining third deletions lower the titer (Fig. 2b). The best strain in the design is 2.05 $\times$, and reaching it requires passing through an intermediate at 0.91 $\times$ on its most forgiving order and 0.52 $\times$ on its least. The simulated campaign is more capable than a real one in every respect that matters, and it still recovers 88% of the achievable titer while stopping two rounds short of the answer.

The greedy trap has a second form worth naming, because it is not the same failure. One of the two monotone routes ends at FKH1, GCN5, MED4 at 1.73 $\times$. A campaign following it would improve at every round and finish with a triple. The simulated campaign does not find it, because at the second round TFC7 (1.80 $\times$) beat FKH1 (1.53 $\times$) and the better local step led to a dead end. Being reachable is necessary for a stepwise campaign to find a combination, and it is not sufficient.

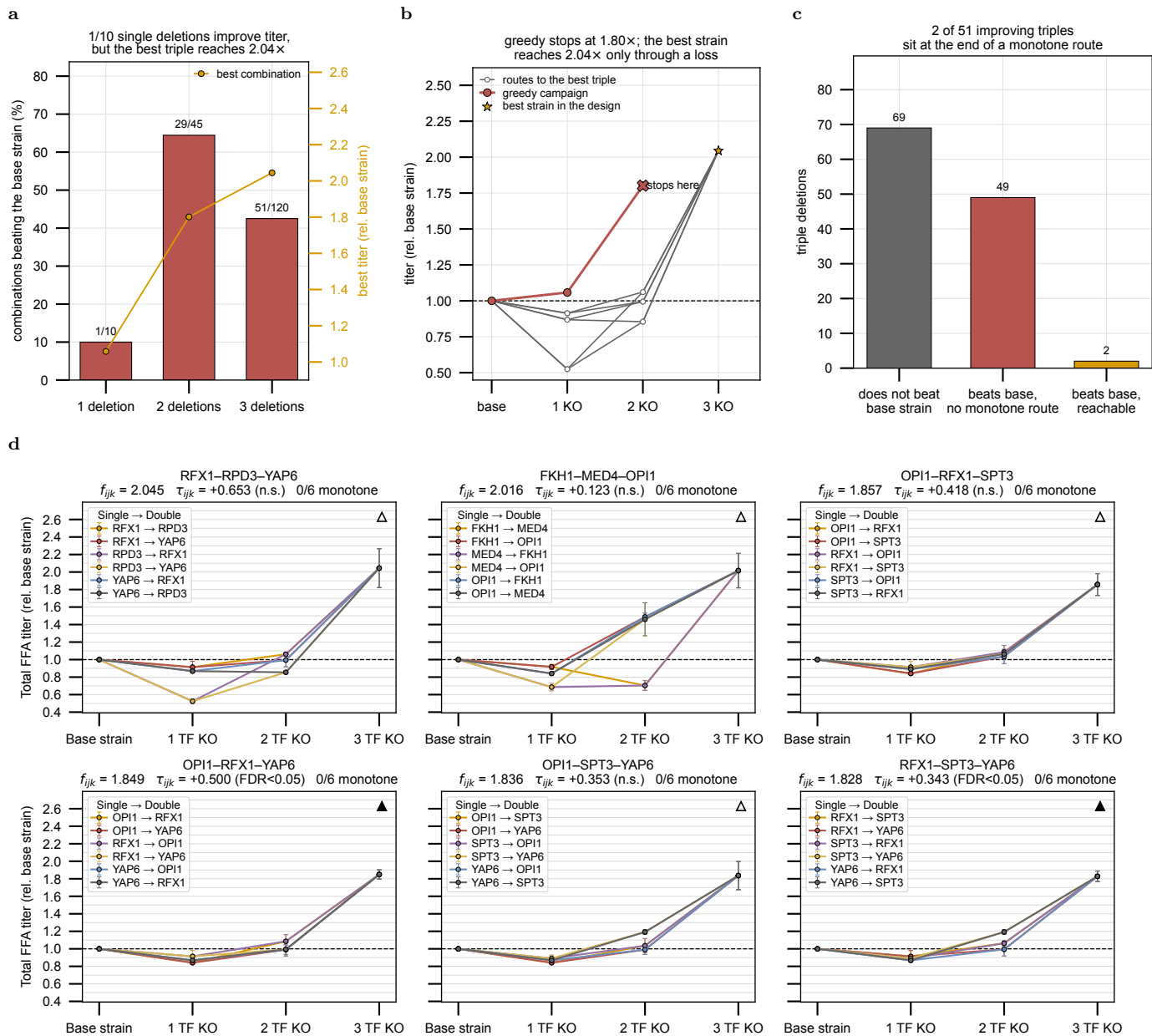


Fig. 2 Improving combinations exist, and almost none of them is reachable one deletion at a time. **a**, Percentage of combinations exceeding the base strain at one, two and three deletions (bars, counts above), with the best titer at each order (amber line, right axis). One of ten single deletions improves titer; the best triple reaches 2.05x. **b**, A greedy campaign (brick) takes the best strictly improving single deletion each round and stops when none improves. It reaches 1.80x after two deletions and halts. Gray lines are all six deletion orders of the highest-titer triple (RFX1, RPD3, YAP6; star), each of which passes through an intermediate below the base strain. **c**, The 120 triples classified by whether they beat the base strain and whether any of their six orders improves at every step. **d**, Trajectories for the six highest-titer triples, one colored line per deletion order, on a shared vertical scale. Rung 0 is the *pox1Δ faa1Δ faa4Δ* chassis and the three rungs after it add transcription factor deletions. Each panel gives that triple's final titer f_{ijk} and its trigenic score τ_{ijk} , and the badge marks the sign and significance of τ . None of the six has a monotone route. Error bars are standard errors of the strain means.

What counts as an interaction depends on the null and on the scale

Equation 1 is one choice among several, and a claim that rests on it should say how much of the claim is the choice. We fit three alternatives on the same normalized strain means: an additive null, in which expected phenotypes add rather than multiply; a generalized linear model with a log link and a Gamma family; and ordinary least squares on log titer differenced against the base strain.

The four models agree on which combinations interact far more than they agree on the sign. Ninety-seven triples are called by at least one model at a 5% false discovery rate within the readout, 73 by all four, and every one of the 73 is negative (Fig. 3a). The disagreement falls almost entirely on the positive calls, and it falls along the scale: the multiplicative and additive models report 11 and 6 positive triples on total titer, and the two log-scale models report none.

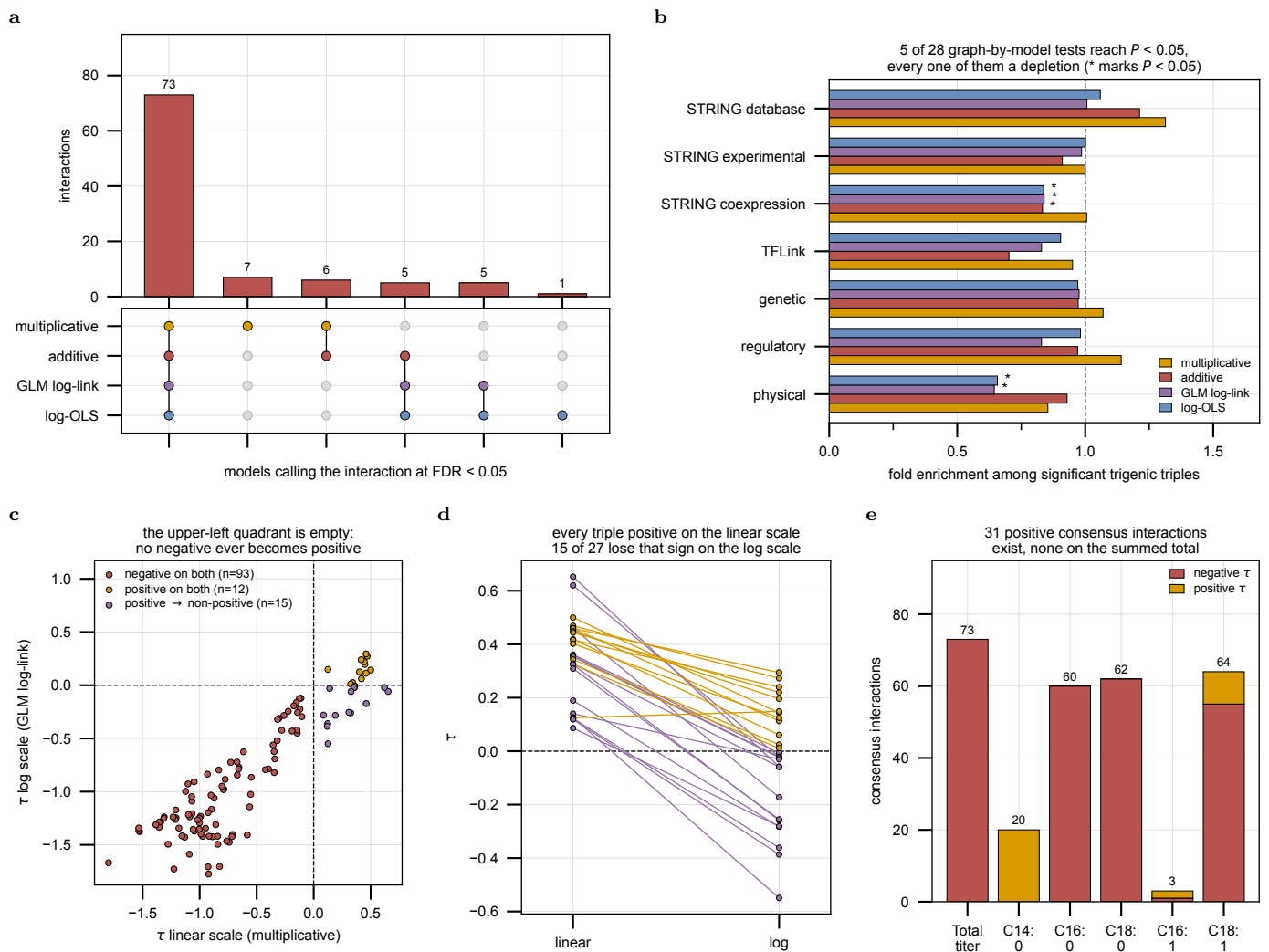


Fig. 3 What counts as an interaction depends on the null, on the scale and on the readout. **a**, Triples called significant at a 5% false discovery rate by each combination of the four models. Ninety-seven triples are called by at least one model and 73 by all four. **b**, Fold enrichment of seven interaction graphs among the significant trigenic triples, per model, over all six readouts pooled at $P < 0.05$. Asterisks mark $P < 0.05$ by Fisher's exact test; all five significant results are depletions. **c**, Trigenic score on the linear scale against the same quantity on the log scale. The upper-left quadrant is empty, so no combination that is negative on one scale is positive on the other. **d**, The 27 triples with a positive linear-scale score, carried onto the log scale; 15 lose the sign. **e**, Consensus interactions per readout, split by sign. Thirty-one positive consensus interactions exist across the five species, and none of them survives on their sum.

That split is not a difference of power, it is a difference of estimand. The log-scale three-way interaction $g_{ijk} - g_{ij} - g_{ik} - g_{jk} + g_i + g_j + g_k$, with $g = \log f$, agrees with the linear-scale score in magnitude ($r = 0.91$, Spearman $\rho = 0.84$) but not near zero. Of the 27 triples with a positive linear-scale score, 15 are non-positive on the log scale, while no negative linear-scale score becomes positive (Fig. 3c,d). Positive three-way epistasis on fatty acid titer is a linear-scale statement, and reporting one without naming the scale is reporting the modeling choice as a result.

The readout matters as much as the scale. Thirty-one triples clear all four models with the sign agreeing on at least one of the five individual species, and 20 of those are on C14:0, where every consensus interaction is positive. None survives on the summed total (Fig. 3e). Summing the five species is a modeling choice too, and it is the choice that removes the positive signal.

No interaction network on hand predicts which combinations interact

If interactions were predictable from prior knowledge, a campaign could avoid the problem by choosing combinations it expects to be additive. We tested that directly by asking whether the interacting triples are enriched for connectivity in existing interaction graphs. Seven graphs cover these ten genes with at least one connected triple: physical interactions, regulatory interactions, genetic interactions, TFLink, and the coexpression, experimental and database channels of STRING 12.0. For each graph and each model we compared the fraction of significant triples whose three genes form a connected subgraph to the same fraction among the non-significant triples.

Nothing is enriched. Across 28 graph-by-model tests, 5 reach $P < 0.05$ and every one of the 5 is a depletion rather than an excess (Fig. 3b). The largest excess anywhere is the STRING database channel under the multiplicative model

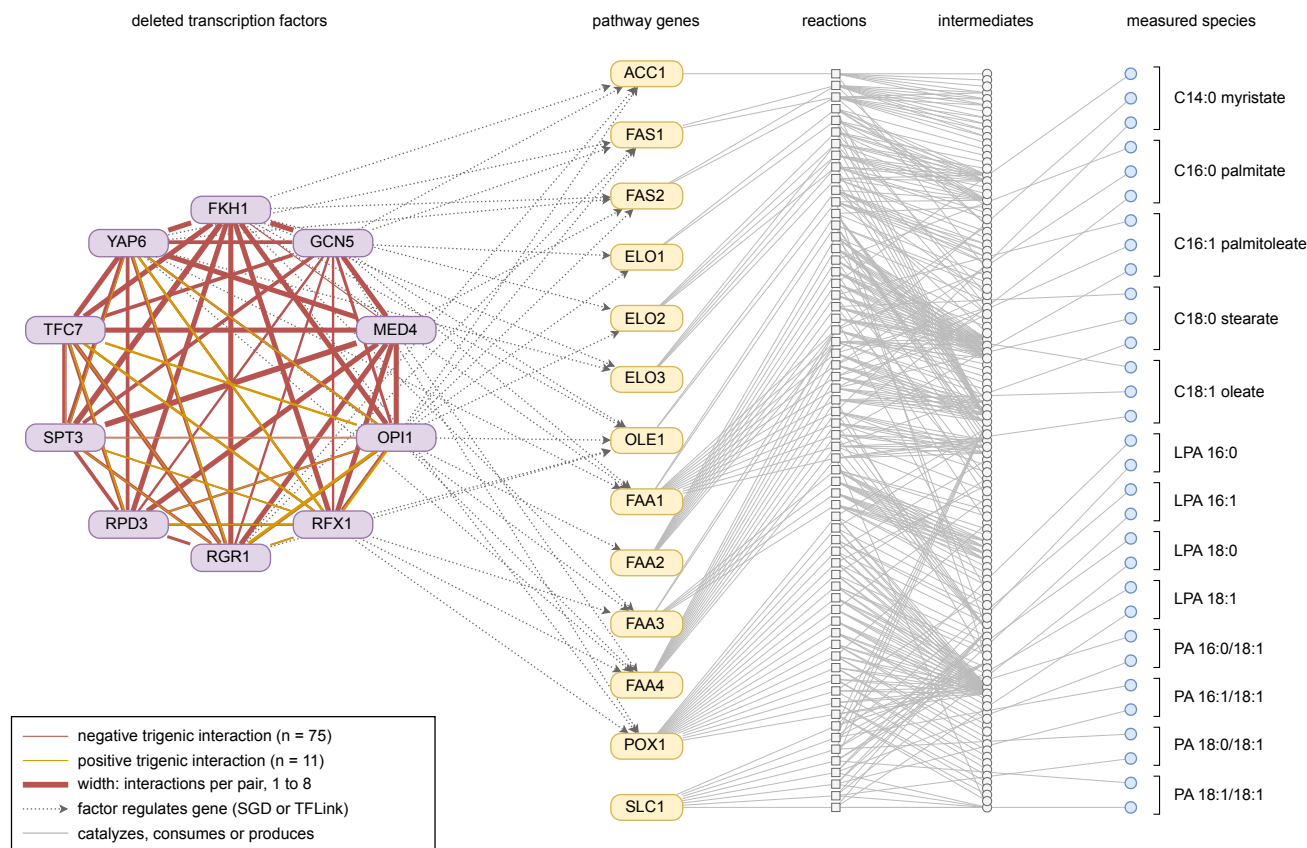


Fig. 4 Trigenic interactions on total fatty acid titer, drawn upstream of the pathway they act through. The ten deleted transcription factors are arranged in a circle at left. Edges within the circle are the 86 trigenic interactions significant at a 5% false discovery rate within the total-titer readout under the multiplicative model, 75 negative (brick) and 11 positive (amber), each drawn as the three edges of its triangle; edge width is the number of interactions the pair takes part in, from 1 to 8, and positive is drawn over negative. Every one of the 45 pairs is in at least one negative interaction. Dotted arrows are regulatory links from a factor to a pathway gene in the SGD regulatory graph or TFLink; three of the ten factors have none. The columns to the right are the 13 core fatty acid pathway genes, the 64 reactions they catalyze, the intermediates those reactions consume or produce, and the 31 measured species, one node per compartment (peroxisome, ER membrane, lipid particle) with a bracket per species. LPA, 1-acyl-sn-glycerol 3-phosphate; PA, phosphatide, written as the two acyl chains.

at 1.31-fold, which does not reach significance ($P = 0.13$). Drawing the significant interactions onto the pathway itself makes the same point visually: the interacting triples span the transcription factor set with no visible relation to which factors regulate which fatty acid genes (Fig. 4).

3 Discussion ✕

The design space of a strain engineering campaign is usually described as too large to search. Ten deletable targets already give 1,023 combinations, and a genome-scale target list gives more combinations than there are strains anyone will ever build. That framing puts the difficulty in the size of the space and implies that the remedy is throughput or a better search.

The measurements here point at a different difficulty, which throughput does not fix. In a space small enough to enumerate completely, with every combination and every construction order measured, the best strain is still unreachable by the procedure that campaigns actually use. The obstacle is not that the good combination is hard to find among many. It is that the good combination sits behind an intermediate that a screening rule would throw away. Of 51 improvements, 2 are reachable. Adding throughput multiplies the number of routes a campaign can try; it does not make a route monotone.

Three consequences follow for how a campaign is designed.

Retain losses. A rule that keeps only strains better than their parent discards the approach to almost every improvement in this design. Carrying a fixed number of non-improving intermediates forward per round costs build capacity in proportion, and it is the only change to the procedure itself that addresses the reachability problem. How many to carry is set by the depth of the valleys, which here have a median of 31% of the base titer.

Build combinations, not sequences. If the intermediates are uninformative about the endpoint, then building them is a cost with no return, and a campaign is better served by constructing the combinations it wants directly. That

is a statement about what multiplexed editing buys: not speed at reaching a known target, but access to targets that no sequence of single edits reaches.

Predict the combination, not the parts. A model that predicts a combination’s phenotype from its parts’ phenotypes fails on exactly the combinations that matter here, because those are the combinations where the parts do not compose. A useful predictor has to be trained on combinations, and the enrichment result says the existing interaction graphs will not stand in for that training data: across 28 graph-by-model tests, the interacting triples are never enriched for connectivity in any graph, and are significantly depleted in five. Whatever determines which combinations interact is not recorded in physical, regulatory, genetic, TFLink or STRING connectivity among these ten genes.

What this design can and cannot support

The design is complete, which is what licenses the reachability claim, and it is small, which bounds it. Ten transcription factors in one producing background, one medium and one harvest time do not establish a rate at which improvements are unreachable in general. They establish that the rate can be 2 in 51 in a real production campaign’s target set, which is enough to make the assumption of composability something that has to be checked rather than assumed.

Two features of the system may not transfer. The deleted genes are transcription factors rather than pathway enzymes, so a deletion acts on many pathway genes at once and the interactions may be denser than a comparable enzyme panel would give. And the phenotype is a titer under production conditions rather than growth rate, so the deletions are under no selection to be individually tolerable. Both cut toward more interaction here than a reader should expect elsewhere. A single-gene-per-step campaign on enzymes may well find a monotone route more often. What the design shows is that the question has an answer, and that the answer is measurable rather than assumable.

What would settle it

Answering the general question needs more datasets of the same shape, and the shape is restrictive. A partial combinatorial screen cannot support a reachability claim at all, because an unbuilt intermediate is indistinguishable from a bad one. The requirement is a design in which every sub-combination of every reported combination was itself built and measured, in the same background and the same batch. Screens that report only the top performers, or only the combinations that a campaign happened to reach, do not qualify no matter how many strains they contain.

The complementary route is theory. A production phenotype is the output of a metabolic network under constraints, and a network model that predicts where in a pathway deletions compose and where they do not would turn the reachability question into a calculation performed before the strains are built. That model does not exist yet for titer, and the enrichment result above says it will not be recovered from connectivity alone. Building it is the direction this line of work points.

4 Methods ✗

Data and normalization

Titers for five fatty acid species and their sum were taken from the source study’s supplementary tables for the base strain and for all 175 deletion combinations. Every titer is divided by the base strain’s titer for the same species in the same experiment, so all values below are ratios and the base strain is 1 by construction. The sum, referred to as total titer, is formed per replicate before averaging rather than by summing the five published species means, so it is a measured quantity and not a sixth derived one. Loading and normalization are in `experiments/008-xue-ffa/scripts/free_fatty_acid_interactions.py`.

Epistasis models

Four nulls were fit on the same normalized strain means.

Multiplicative. The expected phenotype of a combination is the product of its parts. Digenic $\varepsilon_{ij} = f_{ij} - f_i f_j$; trigenic as in Eq. 1. This is the form used by the yeast genetic interaction maps and is the model every figure uses unless it says otherwise (`free_fatty_acid_interactions.py`).

Additive. The same construction with sums in place of products (`additive_free_fatty_acid_interactions.py`).

GLM log-link. A generalized linear model with a Gamma family and a log link, whose three-way coefficient is the interaction (`glm_log_link_epistatic_interactions.py`).

Log-OLS. Ordinary least squares on log titer differenced against the base strain (`log_ols_wt_differencing_epistatic_interactions.py`).

The equivalence used in Fig. 3c,d, that the saturated three-way interaction on log titer reproduces the shipped GLM and log-OLS coefficients, is asserted in code rather than assumed: `linear_vs_log_scale_sign_test.py` recomputes it from the strain means and raises unless it agrees with both fitted models to $r > 0.99$.

Uncertainty and multiple testing

Standard errors propagate through the interaction expression by the delta method, using the per-strain standard errors of the replicate means. The resulting t statistic is referred to the Welch-Satterthwaite effective degrees of freedom of the linear combination, computed from each term's contribution to the combined standard error. This matters more than it usually does. A trigenic score combines seven independently measured strains, and referring its t statistic to the degrees of freedom of the smallest single input, as an earlier version of this analysis did, gives 1 or 2 rather than a median of 4.28, and suppresses nearly every call. Across the 990 multiplicative tests of the whole run, none was significant at a 5% false discovery rate before the correction and 456 are after it (audit record: `notes/plan.008-xue-ffa-epistasis-audit.2026.08.15.md`).

Benjamini-Hochberg controls the false discovery rate. The family matters and is stated wherever a count appears. Counts in Figs. 1 and 3 correct over the tests of a single readout, since that is what those figures claim about. The enrichment analysis in Fig. 3b uses the stored per-test $P < 0.05$ over all six readouts pooled, which is the family the enrichment scripts were written against.

Path enumeration and monotonicity

For each triple, all six orders of introducing its three deletions were expanded into a four-rung trajectory: base strain, one deletion, two deletions, three deletions, taking the measured strain at each rung. A route is monotone when every step is strictly positive, and monotone within one standard error when every step exceeds the negative of that step's propagated standard error. Valley depth is $1 - \min(\text{rungs})$ on a route, and a triple's valley depth is the minimum over its six routes, meaning its shallowest available approach. The greedy campaign in Fig. 2b starts at the base strain, evaluates every combination reachable by one further deletion, takes the highest if it exceeds the current strain, and stops otherwise. Enumeration and monotonicity are in `ffa_epistatic_path_panels.py`; the greedy simulation is in `perspective_figure_panels.py`.

Interaction graph enrichment

Ten interaction graphs were queried for each triple: physical, regulatory and genetic interactions, TFLink, and the neighborhood, fusion, co-occurrence, coexpression, experimental and database channels of STRING 12.0. A triple counts as overlapping a graph when its three genes induce a connected subgraph of that graph. Fold enrichment is the ratio of the overlapping fraction among significant triples to the overlapping fraction among the rest, and significance is Fisher's exact test on the corresponding two-by-two table. Three of the ten graphs contain no connected triple among these ten genes at all and are omitted from Fig. 3b, which is a statement about those graphs' coverage of this gene set and not about epistasis (`interaction_graph_enrichment_analysis.py`).

Figures

Every panel is produced by a committed script and written as a true-size SVG at one of the standard panel widths, then converted to PDF for this document. Panels come from `perspective_figure_panels.py`, `notable_interaction_selection.py` and `ffa_epistatic_path_panels.py`; the network overlay comes from `ffa_network_overlay_panel.py`, and `ffa_network_overlay_drawio.py` writes the same layout as native draw.io shapes, which is the source Fig. 4 is exported from. The multi-panel arrangements are assembled from the panel SVGs by `build_perspective_drawio.py` into editable draw.io sources under `notes/assets/drawio/`.

Code and data availability

All analysis scripts are in `experiments/008-xue-ffa/scripts/` and all derived tables in `experiments/008-xue-ffa/results/` of the torchcell repository. Selection of which interactions appear in a figure is itself a committed artifact

rather than a post-hoc choice: `notable_interaction_selection.py` states the rule, applies it, and writes the selected set to `results/notable_interaction_selection.csv`.